STILLSTACK · USER REQUIREMENTS

StillStack user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First user requirements (kit 1.9.1 upgrade), drawn from the users and stakeholders in SSK-PRB-001

StillStack user requirements

What the households, caretakers and builders of SSK-PRB-001 need from a solar still, written from their side. The product requirements that meet these needs are in SSK-PRS-001 (`03-requirements.md`), and the links are in the generated traceability matrix SSK-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22. Priorities are proposals for review with the first co-design partner.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Enough water from one panel	One panel gives a household of two its drinking and cooking water, about 10 L on a clear day	High	Household (PRB Table 2)
UR-002	Performance	Better than a basin still	The panel makes clearly more water per square metre than the basin still people already know, by reusing the heat it collects	High	Household; NGO (PRB Table 2)
UR-003	Safety	Low-salt distillate	The water that comes out has almost no salt, whatever the feed	High	Household; clinic (PRB Table 2)
UR-004	Safety	Brine never reaches drinking water	Salty water cannot get into the distillate outlet, and the two outlets cannot be mixed up	High	Clinic caretaker (PRB Table 2)
UR-005	Deployment and assembly	No power or fuel	Runs on sunlight alone, filled by hand from a bucket or container	High	Household; outpost (PRB Table 2)
UR-006	Environment and durability	Real feed water	Accepts seawater or brackish well water after a cloth filter, without salt clogging it	High	Island outpost; household (PRB Table 2)
UR-007	Safety	Safe materials	Nothing that touches the distillate adds anything harmful to it	High	Household; health authority (PRB Table 3)
UR-008	Environment and durability	Survives a dry day	Forgetting to fill the feed on a sunny day does not ruin the panel	High	Household; caretaker (PRB Table 2)
UR-009	Logistics	Two people can move it	Two people can carry the panel, and it fits a small yard or roof	Medium	Household; outpost (PRB Table 2)
UR-010	Maintenance	Easy cleaning	One person rinses and refits the wicks without special tools in under half an hour, at most once a week	Medium	Clinic caretaker (PRB Table 2)
UR-011	Usability	Works at my latitude	The panel can be tilted to suit the site, from the tropics to about 35 degrees	Medium	Household; NGO (PRB Table 2)
UR-012	Cost	Affordable and buildable locally	A community workshop can build it from local materials with hand tools and a 3D printer at a cost an NGO can fund	High	NGO, makerspace or college (PRB Table 2)
UR-013	Safety	Stays put and does not burn	The panel stays on its stand in a strong breeze, and the hot parts are guarded by instructions and a cover	High	Household; caretaker (PRB Table 2)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.